

FUNDAMENTALS OF COMPUTER HARDWARE AND CIRCUITS.

1. Compute the average of all the distinct numbers represented by a 5-bit signed binary number.

SOLUTION.

For a 5-bit signed binary number = 8

Then the sum of an arithmetic progression from $-m^{n-1}$ to $m^{n-1}-1$

So, in binary we compute in 2, and the distinct numbers are in 5, so we calculate as

$$-2^{5-1} = -2^4 = -16$$

$$2^{5-1} = 2^4 - 1 = 15$$

So, the arithmetic progression ranges from -16 to 15

The average = sum of numbers counts of numbers

The sum of numbers = -16 and the counts of numbers = $2^5 = 32$

$$\text{So, } -16 \div 32 = -1 \div 8 = -0.5$$

2. Compute the value of $FF6A_{16} - FCD5_{16}$ using a signed magnitude binary number.

SOLUTION.

In hexadecimal, we must convert the values to binary.

Since its 6 values it will be = 24 bits.

So, $FF6A_{16}$ in binary =

$$F = 1111, F = 1111, 6 = 0110, A = 0001, 1 = 0001, 6 = 0110$$

$$\text{So, } FF6A_{16} = 1111\ 1111\ 0110\ 0001\ 0001\ 0110$$

FCD516=

F = 1111, C = 1100, D = 1101, 5 = 0101, 1 = 0001, 6 = 0110

So, FCD516 = 1111 1100 1101 0101 0001 0110

3. Given $x=975_{10}$, $y=891_{10}$, compute $x-y$ using 2's complement Arithmetic.

SOLUTION

Using 2's complement we compute in binary. So, $x = 975_{10}$ we convert to binary =

2	975	Remainder
	487	1
	243	1
	121	1
	60	1
	30	0
	15	0
	7	1
	3	1
	1	1
	0	1

So, $975_{10} = 1111001111_2$

$Y = 891_{10}$

2	891	Remainder
	445	1
	222	1
	111	0
	55	1
	27	1

	13	1
	6	1
	3	0
	1	1
	0	1

$$891_{10} = 1101111011_2$$

Now we convert the bits y to 1's compliments

001101111011

110010000100

Now we add 1 the 1's compliments to get 2's compliment

110010000100

+ 1

=110010000101

Now we add x and the 2's compliments of y to compute x-y using 2's complement arithmetic

-y(2's complement of 891)

-y = 110010000101

+ X= 001111000111

= 1000010001100

4. Given $x=E3BC7_{16}$, $y=AB61_{16}$, compute the value of x-y using 15's complement and present your final answer in octal.

SOLUTION



$Y = 0AB61_{10}$ in





5. Compute the number of additional addresses needed to store an 8-bit unsigned number in an 8-bit signed register.

SOLUTION

So, in 8- bits unsigned

We store values from 0 to 255

Total values: $2^8 = 256$

8- bit signed

We store values from -128 to 127

Total values $2^8 = 256$

So, we store up to 255 but signed 8- bit out at 127

$255 - 127 = 128$

So, there are 128 traditional addresses.

